SPCIFIC OBJECTIVES
By the end of the lesson the learner, should be able to;

a) Definition of PV system sizing
b) State the reasons for sizing a solar PV system
c) Explain factors to be consider when determining the total daily energy demand.
d) Describe how to use basic ERC approved sizing tools and tables.
e) Design solar P.V system for a certain load
f) Analyze how to select the appropriate system voltage

Definition

- Sizing is a process of getting the correct combination of module size, battery size, charge controller, inverter, cables, and other accessories for a solar PV system based on end user needs.

Need/Reason for sizing a solar PV system

1. A Solar PV system is sized in order to determine;
2. The right size of module
3. The right size of charge controller
4. The right size of battery and
5. The right size of cables, fuses and circuit breakers.
6. Proper sizing ensures the best PV-Battery combination to support the end user's daily energy design requirement throughout the year while ensuring suitable battery cycling to maximize battery service life.
7. Oversized systems result in unnecessary additional expense and undersized systems will not regularly deliver the power required leading to batteries being in a consistently low state of charge and consequently shorter battery life and increased operating costs.

Factors to consider when sizing PV systems

End user needs

- It is important that correct information on end user needs is collected outlining required appliances and hours of use.

Losses

- Energy is always lost due to inefficiencies in wire, batteries and inverters. This extra amount should be added to the total daily load energy demand. It is difficult to measure energy losses exactly.
- Very general estimates of energy losses can be made as follow:

- If the system components are new and properly sized, estimate energy losses to be 15% of estimated system energy Daily load.
- If there are a number of long wires runs (over 10meters), and the equipment is new, estimate energy losses to be 20% of estimated system energy daily load.
- If the system uses an inverter, estimate system losses to be 30% of estimated system energy daily load

Insolation

- This refers to the average sun hours falling on a particular location. Insolation data is available from Meteorological department.
- Insolation maps could be used to determine insolation of a particular location. (Ref. Solar Electric Systems for Africa Page 8&9).
- **5 sun hours** global insolation is usually acceptable when sizing.
- Actual insolation data can be found through the following website: http://re.jrc.ec.europa.eu/pvgis/apps4/pvest.php?map=africa&lang=en (KEREA Technical Manual Page 29)

Days of autonomy

- This refers to the number of days that a fully charged battery/battery bank can supply energy to the system loads when there is no energy supplied by the PV array. For common, less critical PV applications, autonomy periods are typically between two and four days.

Acceptable daily depth of discharge (DDOD)

- The acceptable DDOD for modified automotive solar/TV batteries is 50%, for sealed solar batteries is 60% and for flooded deep cycle solar batteries is 80%.

PV System sizing/design using simplified method for DC systems only (T1)

- This is a two-step process through which one can reasonably determine the size of module(s), battery bank and charge controller required for given daily energy requirements.
- PV systems are utilized based on the type of electric appliances used by the users. This information is gathered through a site survey. A total energy requirement per day of (watt-hour per day) is estimated by using the calculation sheet.

Step 1 Determining Daily energy requirements

- The daily energy requirements is determined by filling the table below in consultation with the customer by entering the type and number of appliance and the number of hours of use per day.

- The daily energy requirement of each appliance is calculated by multiplying the appliance rating (watts) by the number of hours of use.
- The total daily energy requirement is the sum of energy requirement for all the appliances.

Determining Daily energy requirements

Appliance	No.	Nominal Voltage(V)	Wattage Rating (W)	Daily Hrs. of use	Daily Energy consumption (Watt-hrs.)
D.C 14” colour T.V	1	12	55	3	165
Indoor lights	4	12	8	3	96
Outdoor light	1	12	9	1	9
Radio	1	12	10	2	20
Phone charger	1	12	5	2	10
Total					300
Total Daily Energy Demand (Wh) 300					

Step 2

- Using system design table on page 27 KEREA Technical manual, chose the nearest value in the first column to the total obtained in step 1, and then read across the column to get the recommended size of module/array, battery bank and charge controller.
- Note that the table is based on the following assumptions:
  - ✓ 20% system loses
  - ✓ Five peak sun shine hours
  - ✓ Battery with 50% allowable depth of discharge
  - ✓ Battery bank sized to store 3 days energy (days of autonomy)
- **NB:** When using an inverter an additional 10% for inverter losses
- From the system design table, the recommended sizes of module, battery and charge controller are as follows:
- Module = 105W
- Battery = 130Ah
- Charge controller = 10A

Class Exercise

- Mr. Mwazera requires a solar PV system to power the following 12V DC appliances in the table below;

Appliance	No.	Nominal Voltage(V)	Wattage Rating (W)	Daily Hrs. of use	Daily Energy consumption (Watt-hrs.)
Indoor lights	5	12VDC	9	4	
Outdoor lights	2	12VDC	7	2	
DC black & white T.V	1	12VDC	20	4	

DC Stereo system	1	12VDC	40	3	
Total					
Total Daily Energy Demand (Wh)					

Using KEREA system design table, determine the following;

i. Total Daily Energy Demand
ii. Correct module size
iii. Correct charge controller size and
iv. Correct battery size

NOTE: EXTRACTED DESIGN TABLE FROM KEREA TECHNICAL MANUAL ON PAGE 27

STEP 2: Using the nearest value in the first column to the total obtained in step 1, read across the column to get the recommended size of module/array, battery bank and charge controller

System Design Table

Daily Energy Req. (Wh)	Daily Energy Req. (Wh) (considering 20% losses)	Daily Energy Req. (Ah)	Module Current (Amps) (considering 5psh)	Module Array (Wp)	Battery bank (Ah) (50% DOD and 3 storage days)	Charge controller (Amps)
40	48	4.0	0.8	14	26	5
60	72	6.0	1.2	20	26	5
80	96	8.0	1.6	30	26	5
100	120	10.0	2.0	40	40	5
120	144	12.0	2.4	45	50	5
140	168	14.0	2.8	50	50	5
160	192	16.0	3.2	50	65	5
180	216	18.0	3.6	65	75	5
200	240	20.0	4.0	65	75	5
220	264	22.0	4.4	75	100	10
240	288	24.0	4.8	80	100	10
260	312	26.0	5.2	85	100	10
280	336	28.0	5.6	105	100	10
300	360	30.0	6.0	105	130	10
☛ 320	384	32.0	6.4	110	130	10
340	408	34.0	6.8	115	130	10
360	432	36.0	7.2	120	150	10
380	456	38.0	7.6	125	150	10
400	480	40.0	8.0	140	150	10
420	504	42.0	8.4	140	150	10

This table is based on the following assumptions.

- 20% system loses
- Five peak sun shine hours
- Battery with 50% allowable depth of discharge
- Battery bank sized to store 3 days energy

NB: When using an inverter an additional 10% for inverter lossess